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Graded Assignment • 20 min

Due Mar 11, 11:59 PM CET

1. The amount of light collected by a telescope on a given celestial object is determined by which of the following: 1 point

- ○ the size of the detector
- ○ the thickness of the primary mirror
- ○ the area of the secondary mirror
- ◉ the area of the primary mirror

2. The smallest angular size that the human eye can resolve is: 1 point

- ○ 30 arcminutes
- ○ 25 arcseconds
- ◉ 1 arcminute
- ○ 1 arcsec

3. Which two techniques most effectively correct for blurring in astronomical images due to the atmosphere? 1 point

- ☐ only observe very bright targets
- ☑ put your telescope in space
- ☑ observe a bright star with very, very short exposures and use that to correct for the atmosphere
- ☐ only observe on the nights with the best weather and from the best sites.
- ☐ only observe targets when they are very close to zenith, i.e. when they are highest in the sky

4. How are stars distributed on a colour-magnitude diagram (a.k.a. HR diagram)? 1 point

- ○ The colors and luminosities of stars are mostly similar to that of the Sun.
- ○ The Sun is one of the most massive stars. Most stars are fainter than the Sun, so they appear lower on the diagram compared to the Sun.
- ○ The Sun is a very low mass star and thus faint. Most stars are brighter than the Sun, so they appear higher up on the diagram compared to the sun.
- ○ The colors and luminosities of stars are randomly distributed
- ◉ Most stars on the HR diagram would lie along the main sequence, with bright blue stars at the top of the diagram and dimmer red stars towards the bottom and right.

5. Stars are very massive, so gravity should force them to collapse. So, what holds up a main sequence star and prevents it from collapsing? 1 point

- ○ The energy produced by fusing helium atoms together produces pressure which counteracts gravity.
- ○ Neutron degeneracy in the core of the star prevents the star from collapsing.
- ○ Electron degeneracy in the core of the star prevents the star from collapsing.
- ○ The energy produced by fission of iron nuclei produces pressure which counteracts gravity.
- ◉ The energy produced by fusing hydrogen atoms together produces pressure which counteracts gravity.